

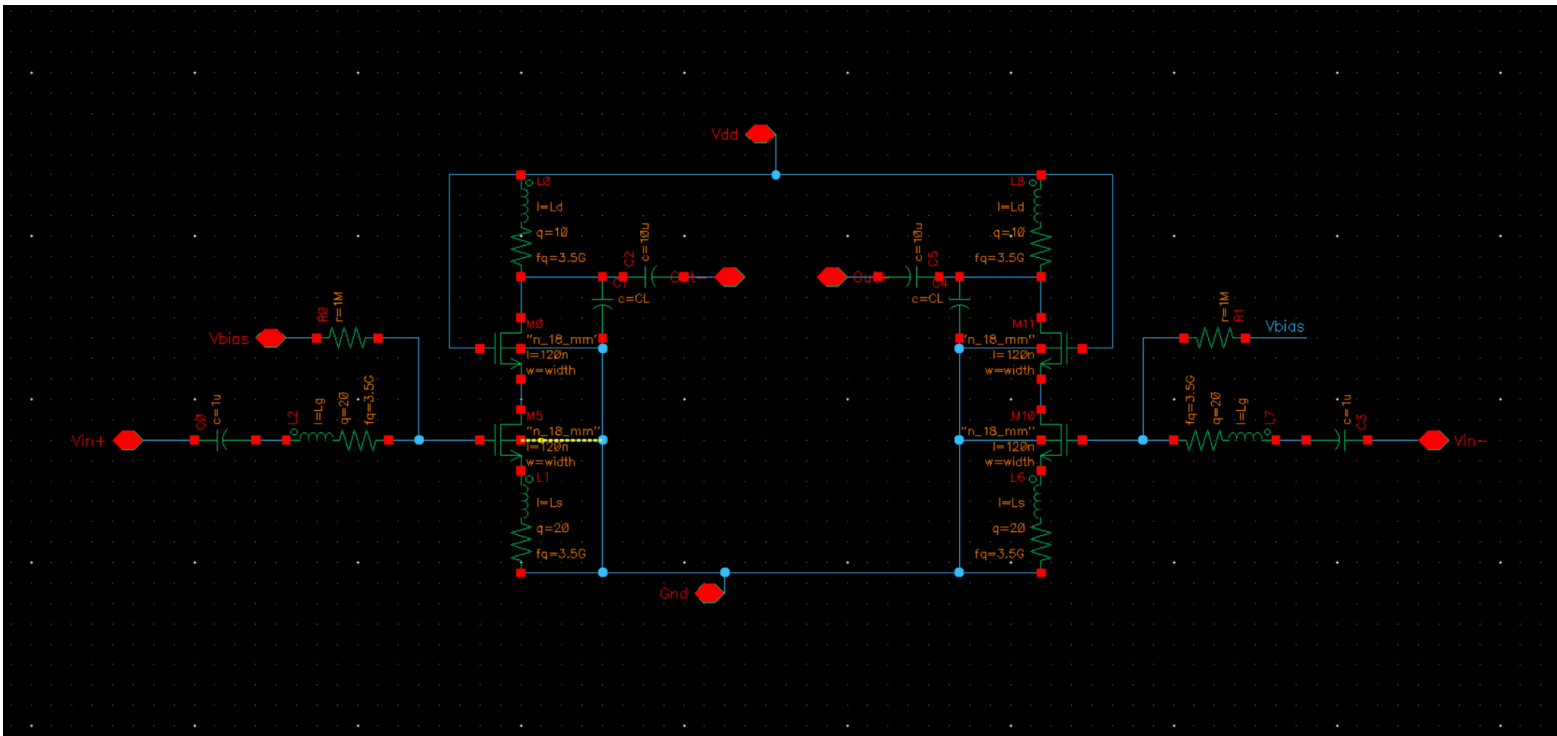
EE6320 RF Integrated Circuits

Project: LNA Design

Performance Summary Table			
Design metric	Measurement	Simulation Result	Requirement
Input matching	Worst case S11 in the specified band	-11.56	< -10dB
	Band over which S11 $\leq$ -10dB	3423 to 3656 MHz	3450 to 3550 MHz
Voltage Gain	Minimum Gain in the specified band	36.09 dB	$\geq$ 20dB
	Maximum Gain in the specified band	36.56 dB	$\geq$ 20dB
	Gain flatness in specified band [Max-Min Gain]	0.47 dB	$\leq$ 1dB
	3dB Bandwidth	0.331 GHz	-
	Load Capacitance [Differential]	1p F	200f F
Noise Figure	Maximum Noise Figure in the specified band	1.80 dB	$\leq$ 2dB
	Minimum Noise Figure in the specified band	1.77 dB	-
	Band over which NF $\leq$ 2dB	3.05GHz to 4.45GHz	-
Linearity	IIP3 Tones used	3.5GHz, 3.501 GHz	-
	Input power used for extrapolation	-40 dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-3.36 dBm	-
	Power of IM3 Tone at output (at chosen input power)	-49.19 dBm	-
	Extrapolated IIP3	-9.523 dBm	$\geq$ -10 dBm
Power	LNA DC power consumption [Excluding Bias]	2.456 mW	Minimize
	Bias circuit power consumption (power supplied by i_bias)	1.086 mW	Minimise
Other	Sum of all on-chip inductances	4.38 nH	-
	Sum of all off-chip inductances (2 Lg's)	38 nH	-
	Sum of all resistances [Including bias]	2 M Ω	-
	Sum of all capacitances [Including AC coupling, excluding load]	22 μ F	-
	Simulator Used	Cadence Spectre	-

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LNA Schematic



Component Values (one side values)

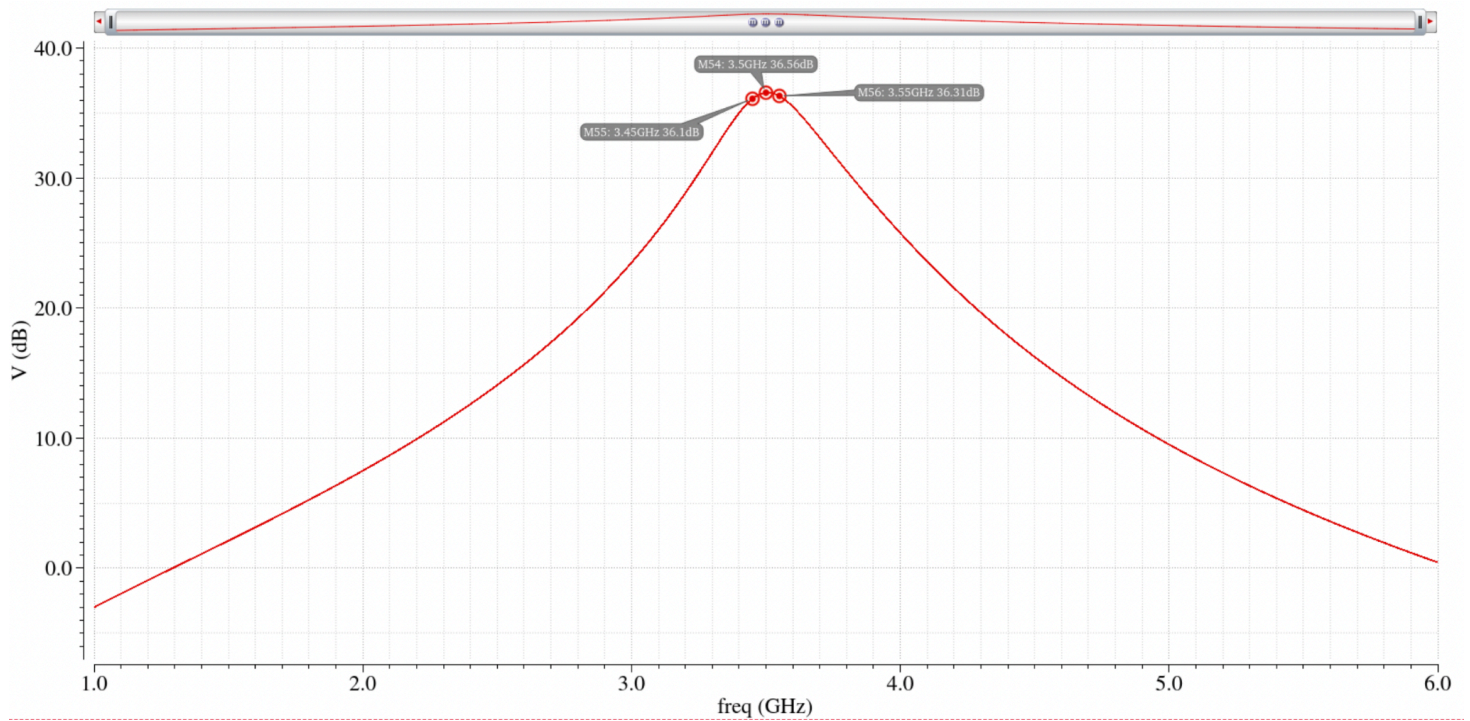
<u>Design Component</u>	<u>Hand Calculated Value</u>	<u>Simulated Value</u>
Lg (gate inductance)	295 nH	19 nH
Ls (source inductance)	130 pH	220 pH
Ld (drain inductance)	2.06 nH	1.97 nH
Rd* (drain resistance)	457 Ω	437 Ω
R _{Is} * (res. parallel with Ls)	14 Ω	24 Ω
Rg* (res. parallel with Lg)	32 k Ω	2 k Ω

Fixed Constant Parameters

- LNA MOS parameters: W = 50 μm , L = 0.12 μm
- Current Mirror MOS parameters: W = 100 μm , L = 0.12 μm
- I_{bias} (current): 2 mA
- C_{coupling}: 1 μF

* - intrinsic resistance to respective inductors (doesn't come in net resistance on/off the chip)

Gain Plot (from AC)



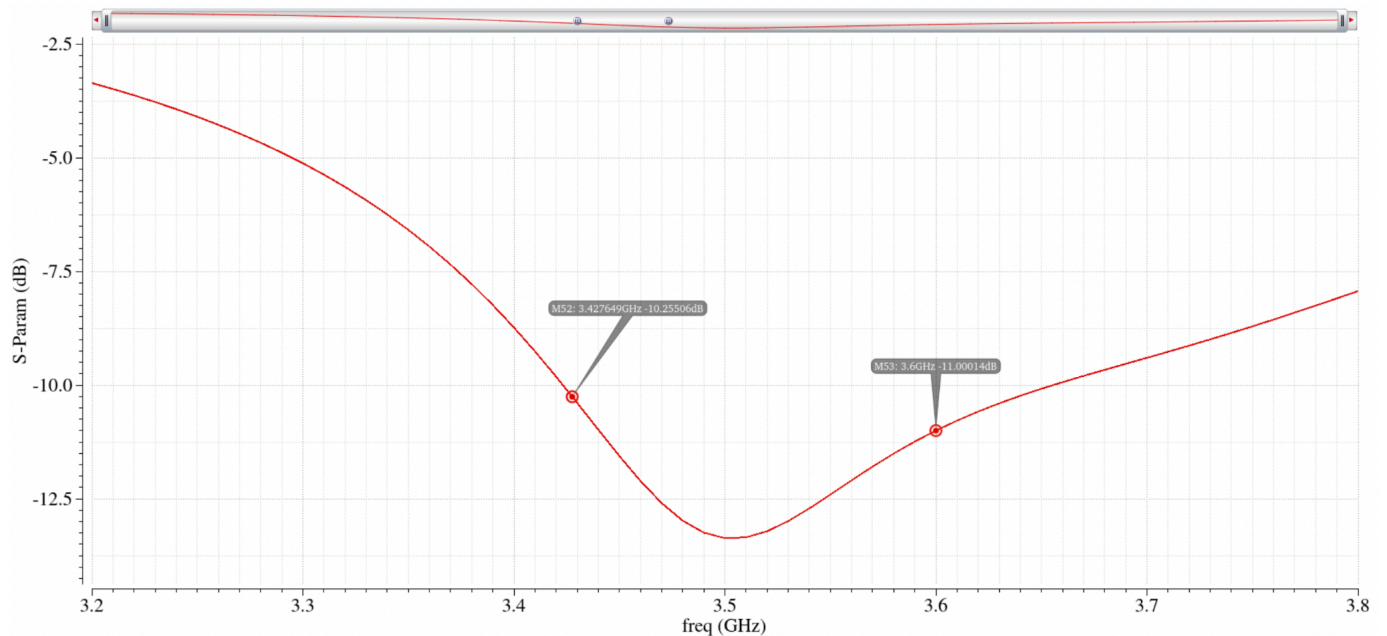
Max Gain through hand calculation: 45 dB

Max Gain from the simulation: 36.56 dB

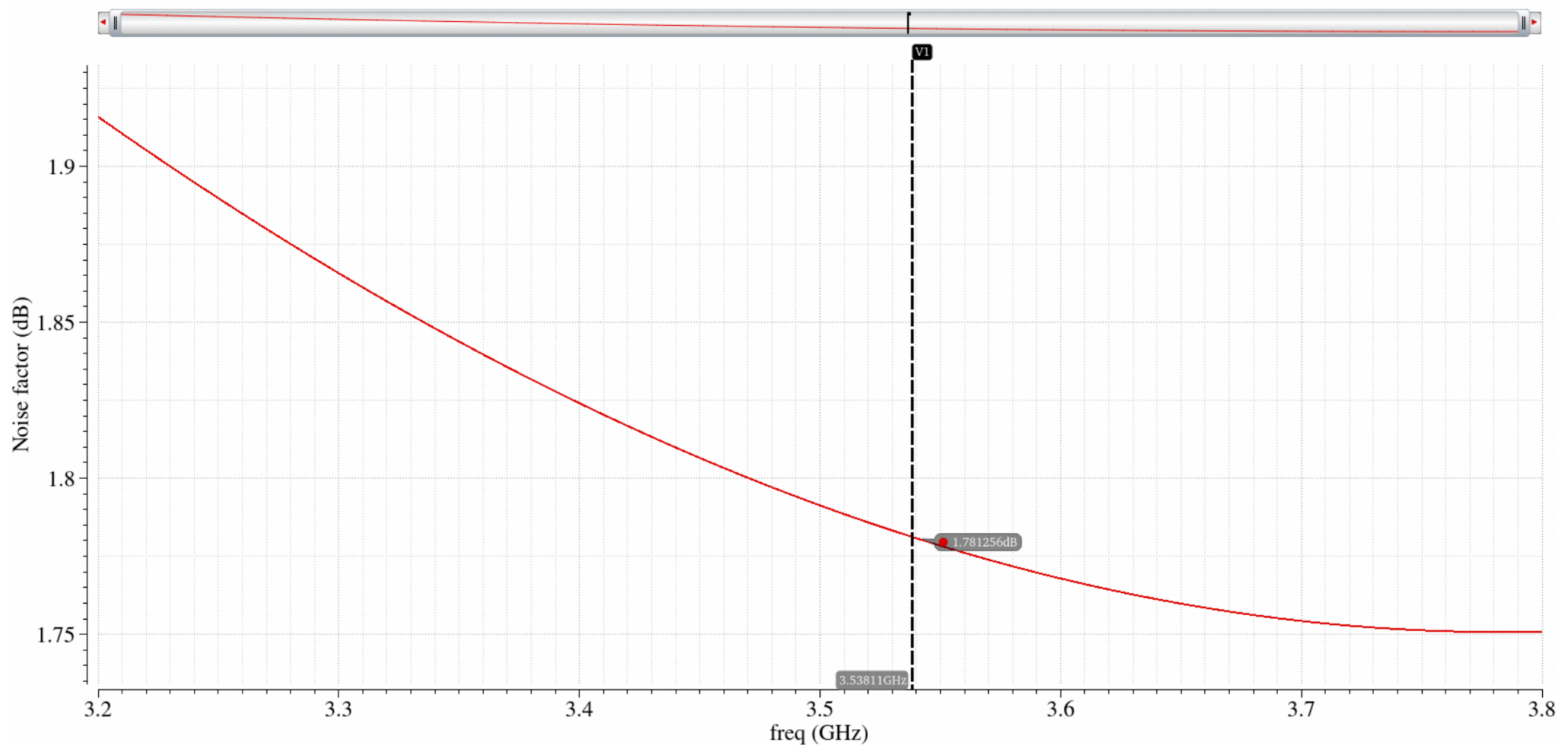
-3dB Gain comes out to be: 33.56 dB

From the figure, 3dB bandwidth comes out to be: 0.332 GHz

S11 Plot



NF Plot

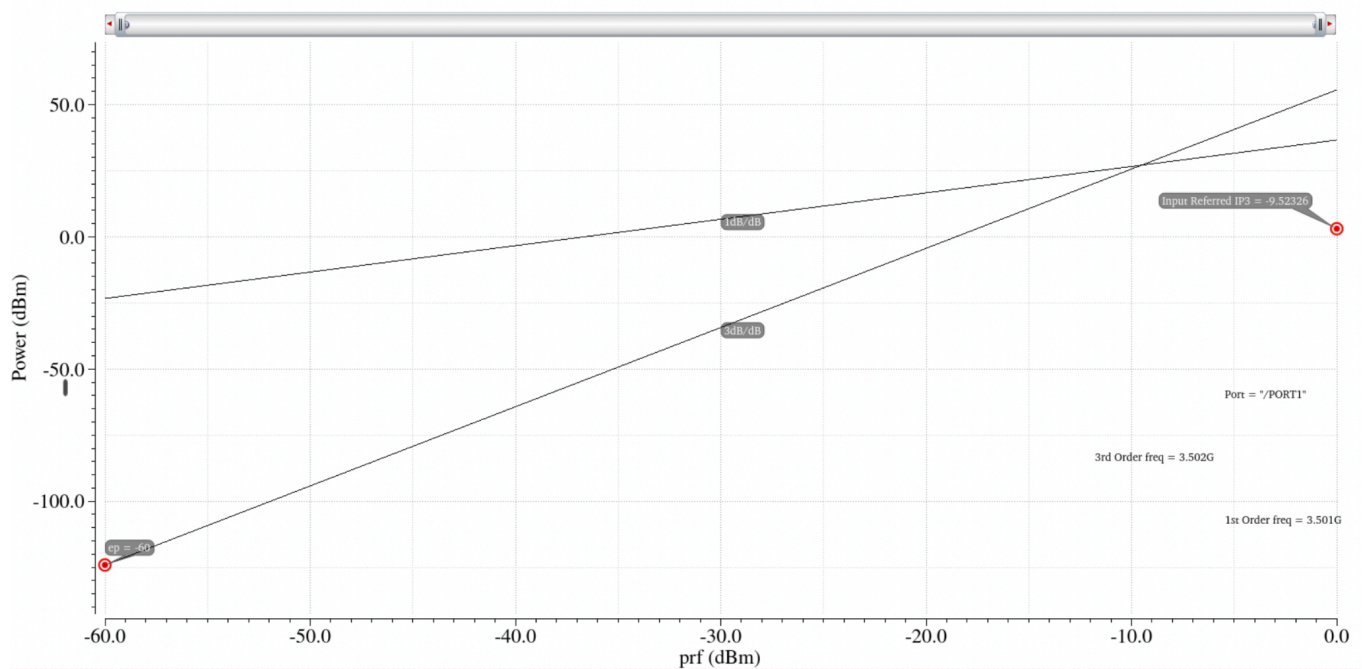


From hand calculations, NF comes out to be: 0.885 dB

From the plot, NF comes out to be: 1.79 dB

Linearity Plot (IIP3 Computation) (only for representation)

Sweep power from very low powers to very high power and show the extrapolated point

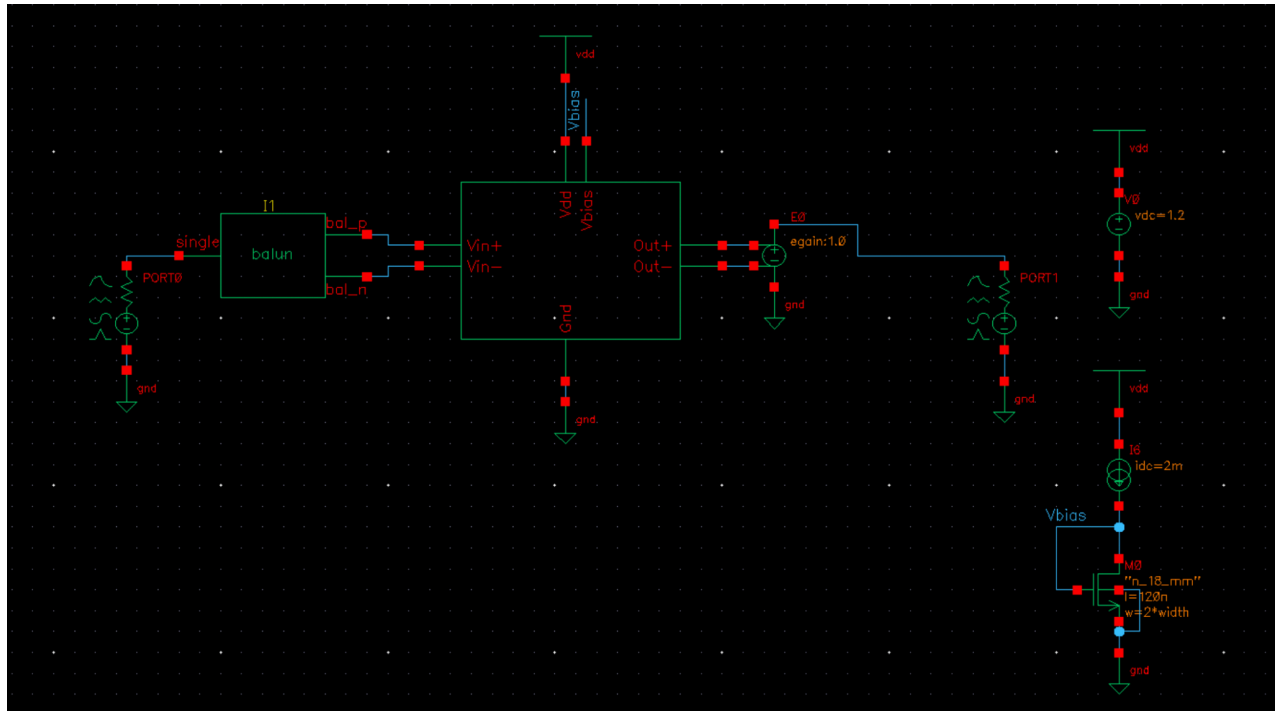


Tones used: 5.5 GHz and 5.501 GHz

IIP3 point comes out to be: -9.523 dBm

LNA Testbench

This testbench is used to compute S11, NF, and IIP3 point.



Comments

- The inductive source degeneration technique was effective in achieving input matching to 50 Ω . By tuning L_s and L_g , the imaginary part of the input impedance was cancelled at the design frequency of 3.5 GHz, resulting in a minimum S11 around this frequency. However, the parasitic resistances of the inductors had to be taken into account to obtain good agreement between the calculated values and the simulated design.
- The cascoding transistor significantly improved gain by increasing the effective output resistance and reducing the Miller effect. This also improved reverse isolation and stability.
- The gain exhibits a peak near the resonance frequency due to the LC load network. Away from resonance, the gain decreases as the impedance of the tank circuit reduces.
- Increasing the width of the input transistor increased g_m and improved gain, but also increased C_{gs} , which shifted the resonance frequency and required retuning of the input matching network.
- Increasing the bias current increased transconductance (g_m) and gain but also increased power consumption. The bias point was chosen as a trade-off between gain and power efficiency and also the IIP3 point (non-linearity).
- Since the input transistor dominates the noise performance, its size and bias current significantly affect the noise figure. Higher g_m reduces noise but increases power consumption.
- The cascode structure improves stability by reducing feedback through C_{gd} and improving reverse isolation.

Hand Calculations done for the Project

① LNA

Req:-

$$V_{dd} = 1.2V$$

$$L_{min} = 120nm$$

$$W_{min} = 160nm$$

$$f \Rightarrow 3.5GHz \rightarrow B.W. = 100MHz$$

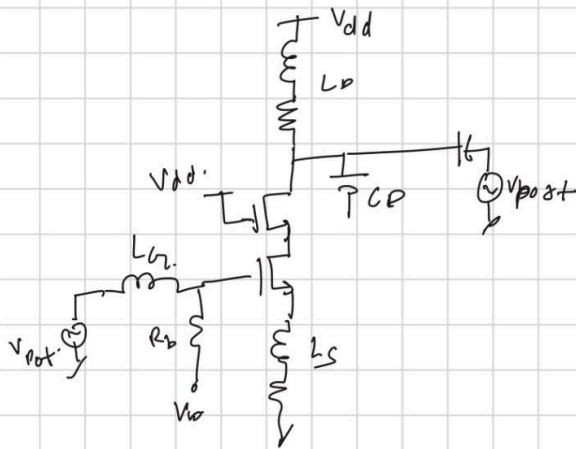
$$C_L = 200fF$$

$$\text{Gain flatness} \leq 1dB \rightarrow Q$$

$$NF \leq 2dB$$

$$IP_3 \geq -10dBm$$

$$\text{Voltage Gain} \geq 20dB$$



$$F = 1 + 2g_m R_S \left(\frac{\omega_0}{\omega_T} \right)^2 + \frac{4R_S}{R_D} \left(\frac{\omega_0}{\omega_T} \right)^2$$

for small F we need $\omega_0 \ll \omega_T$

$$\frac{V_o}{V_s} = -g_m R_D Q_1 \quad \left| \quad \begin{aligned} & \omega_T = \omega_T L_S \\ & Q_1 = \frac{1}{2\omega_0 C_{gs} R_S} \end{aligned} \right.$$

$$\omega_T = \frac{g_m}{C_{gs}} \quad \left| \quad Q_1 = \frac{1}{2\omega_0 C_{gs} R_S}$$

$$f_0 = \frac{1}{2\pi \sqrt{L_D (C_L + C_{db})}}$$

$\hookrightarrow$ very less $\hookrightarrow 200fF$

$$L_D = \frac{1}{C_L (f_0 2\pi)^2} = \frac{1}{200 \times 10^{-15} (3.5 \times 10^9 2\pi)^2}$$

$$L_D = 10.34nH \approx 10nH$$

$$Q = 20$$

$$L_D = 10nH$$

$$Q = \frac{\omega L_D}{R_D} \quad \left| \quad R_D = \frac{2\pi f_0 L_D}{Q} = \frac{2\pi \times 3.5 \times 10^9 \times 10 \times 10^{-9}}{20}$$

$$R_D = 11\Omega$$

$$\text{given gain} \geq 20dB \Rightarrow 10$$

$$\frac{V_o}{V_i} = -\frac{1}{2} \frac{R_D}{R_S} \frac{\omega_T}{\omega_0} = 10$$

$$-\frac{1}{2} \frac{11}{50} \frac{\omega_T}{3.5 \times 10^9} \geq 10$$

$$\omega_T \geq \frac{10^3 \times 3.5 \times 10^9}{11}$$

$$\omega_T \geq 318GHz$$

$$L_D = 7.26$$

$$L_R = 7.$$

$$IP_3 \propto g_m \propto I_D$$

$$W = 10\mu m$$

$$g_m = 6.5mS$$

$$\omega_T = 292GHz$$

$$(\pi f_0)^2 = \frac{1}{L_D C_D}$$

$$C_D = \frac{1}{7.26 \times 10^{-9} \times 3.5 \times 10^9 \times 2\pi^2} \times 3.5 \times 10^9$$

$$R_D = 15$$

$$L_D = 7.26$$

$$Q = 20 = \frac{\omega L_S}{R_p}$$

$$\omega L_p = 13 \times 20 = 160$$

$$L_p 2\pi \times 3.5 \times 10^9 = 160$$

$$L_p = \frac{160}{2\pi \times 3.5}$$

Design Procedure

- The design was initially started by calculating the approximate values of the required components using the design methodology discussed in the lectures. Based on these calculations, the initial values of the transistor dimensions and passive components were obtained.
- These calculated values were then implemented in the circuit simulator. However, due to device parasitics and non-idealities, the simulated results did not exactly match the theoretical calculations. Therefore, iterative tuning of the circuit parameters was performed to achieve the required specifications.
- During the tuning process, the insights and design intuitions developed through the lectures were used to understand how different parameters affect the circuit performance. In particular, the values of the inductors and the transistor dimensions were adjusted to achieve proper input matching and the desired gain at the design frequency.
- Additionally, the parasitic effects of inductors and device capacitances were taken into account to obtain more realistic results. Through multiple simulation iterations, the component values were refined until the circuit met the desired performance near the target frequency of **3.5 GHz**.